Unified Mentor Project

Product Line Profitability & Margin Performance Analysis

Nassau Candy Distributor

Research Paper: Exploratory Data Analysis, Insights and Recommendations

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Data period: 2 January 2024 - 31 December 2025 | 10,194 order lines

Abstract

Sales volume alone is a misleading measure of product success. This paper analyses 10,194 order lines (January 2024 to December 2025) from Nassau Candy Distributor to determine which product lines actually generate profit, which divisions dilute margin, and where pricing, sourcing or portfolio decisions are needed. The business earned $141,784 in sales and $93,443 in gross profit, a portfolio gross margin of 65.9%.

The analysis shows an extremely concentrated business. Five Wonka Bar products account for 92.9% of revenue and 95.1% of gross profit, and only 5 of the 15 products (33%) are needed to reach 80% of both revenue and profit. Two products represent clear margin risk: Kazookles earns a gross margin of only 7.7% (cost is 92% of price), and Lickable Wallpaper, the only non-chocolate product with meaningful sales, earns 50.0%, well below the 66.5% to 71.4% of the chocolate bars. Seven Sugar-division products and Wonka Gum together contribute under 1% of revenue and are candidates for portfolio review.

A further structural finding is that unit price and unit cost are constant for every product across all 24 months, so margin volatility is effectively zero. Portfolio margin is therefore determined by product mix, not by discounting or cost drift. Fixing the two margin-risk products alone, at constant volume, would raise gross profit by roughly $1,417 (+1.5%); the larger strategic exposure is dependence on one division and one factory.

1. Introduction

For a confectionery distributor, high sales do not guarantee high profit. Some products sell in volume but earn thin margins; others consume a disproportionate share of manufacturing cost. Without product-level visibility, decisions on pricing, promotion and portfolio remain reactive and intuition-based.

1.1 Problem statement

The organisation lacks clear answers to four questions: which product lines deliver the highest gross margin; whether high-sales products are actually profitable; how profitability varies across divisions; and which products represent margin risk.

1.2 Objectives

- Calculate gross margin, profit per unit and profit/revenue contribution for every product.
- Rank products and divisions, and segment products by sales level and margin level.
- Measure profit concentration using Pareto analysis and dependency indicators.
- Diagnose cost structure and flag products for repricing, cost renegotiation or discontinuation review.
- Deliver the findings as this paper, an interactive Streamlit dashboard and an executive summary.

2. Data and Cleaning

2.1 Dataset

The dataset contains one row per order line with 18 fields: order and ship dates, ship mode, customer geography (country, state/province, city, postal code, region), division, product ID and name, sales, units, cost and gross profit. It covers 15 products in 3 divisions (Chocolate, Sugar, Other), 8,549 distinct orders, 4 regions and 59 states/provinces (9,994 U.S. rows and 200 Canadian

rows). The brief also supplies a product-to-factory mapping for five factories, which was joined to every row.

2.2 Validation steps and results

Check	Result
Missing values (all fields)	None
Zero or negative sales / units / cost	0 rows
Gross Profit = Sales - Cost (tolerance $0.02)	Holds on all 10,194 rows
Duplicate Row IDs	0
Order dates parsed (dd-mm-yyyy)	100% valid, 2 Jan 2024 - 31 Dec 2025
Product label standardisation	'Wonka Bar -Scrumdiddlyumptious' corrected to 'Wonka Bar - Scrumdiddlyumptious'; labels trimmed
Product ID to product name	One-to-one for all 15 products
Ship Date anomaly	All ship dates fall 904-1,642 days (median 1,274) after order date, beyond Dec 2025

No rows needed to be removed. The ship-date anomaly (ship dates lying years after order dates) looks like a data-generation or date-shift artefact. Because this study concerns profitability, not delivery performance, Ship Date was excluded from all calculations, but the issue should be corrected at source before any logistics analysis.

3. Methodology

3.1 Metrics

KPI	Definition
Gross Margin (%)	Sum of Gross Profit / Sum of Sales x 100 (aggregated first, never an average of row margins)
Profit per Unit	Gross Profit / Units
Revenue Contribution (%)	Product sales / total sales
Profit Contribution (%)	Product gross profit / total gross profit
Margin Volatility	Standard deviation of monthly gross margin (percentage points); requires at least 3 months of sales
Cost Ratio (%)	Cost / Sales; the complement of gross margin
HHI	Herfindahl-Hirschman index of profit shares (0 to 10,000); above 2,500 indicates high concentration

3.2 Segmentation and risk-flag rules

Products were placed in four segments using two cut-offs: high-sales means at least 5% of total revenue, and high-margin means at least the margin threshold (default 60%, adjustable in the dashboard). Risk flags follow explicit rules:

- **Critical:** gross margin below 25%, so reprice and renegotiate cost.
- **Below target:** margin below threshold with at least 1% revenue share, so cost renegotiation.
- **Discontinuation review:** margin below threshold and revenue share below 0.5%.
- **Low-volume watch:** healthy margin but revenue share below 0.5%.

All calculations are implemented in a single Python module (src/metrics.py) shared by the analysis script and the dashboard, so figures in this paper and the dashboard reconcile exactly.

4. Exploratory Data Analysis and Results

4.1 Overall performance

The business generated $141,784 in sales, $48,341 in cost and $93,443 in gross profit from 38,654 units, a gross margin of 65.9% and profit of $2.42 per unit. Sales grew 44.6% from $57,956 in 2024 to $83,827 in 2025 while gross margin stayed flat (65.8% to 66.0%), so growth came from volume rather than from any change in unit economics.

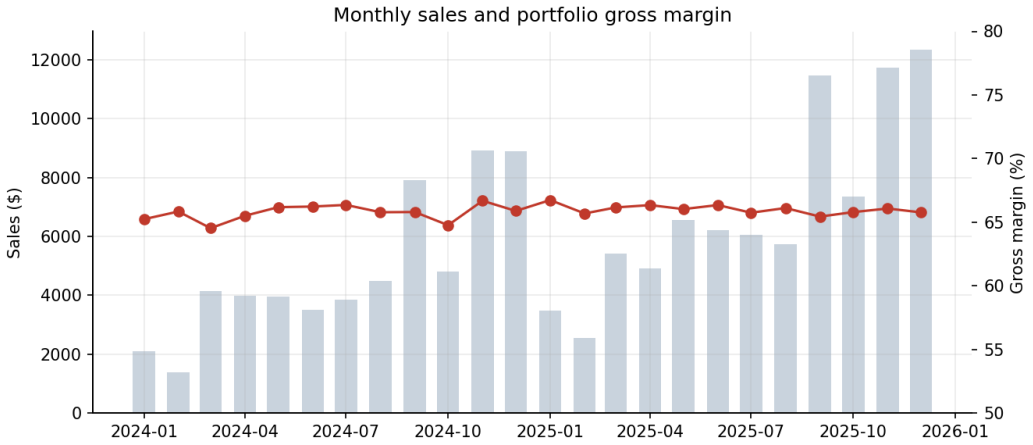


Figure 1. Monthly sales and portfolio gross margin

4.2 Product-level profitability

Table 1 ranks all 15 products by gross profit. The five Wonka Bars occupy ranks 1 to 5, each contributing 17.8% to 20.7% of profit. Nutty Crunch Surprise has the highest margin (71.4%) and Scrumdiddlyumptious the highest profit ($19,358). Milk Chocolate has the lowest bar margin (64.9%) but still sits close to the portfolio average.

#	Product	Division	Sales $	Profit $	GM %	Profit/Unit $	Rev %	Profit %
1	WB: Scrumdiddlyumptious	Chocolate	27,875	19,358	69.4	2.50	19.66	20.72
2	WB: Triple Dazzle Caramel	Chocolate	28,485	18,610	65.3	2.45	20.09	19.92
3	WB: Milk Chocolate	Chocolate	26,868	17,443	64.9	2.11	18.95	18.67
4	WB: Nutty Crunch Surprise	Chocolate	23,575	16,820	71.3	2.49	16.63	18.00
5	WB: Fudge Mallows	Chocolate	24,890	16,594	66.7	2.40	17.56	17.76
6	Lickable Wallpaper	Other	7,860	3,930	50.0	10.00	5.54	4.21
7	Wonka Gum	Other	598	311	52.0	0.65	0.42	0.33
8	Everlasting Gobstopper	Sugar	130	104	80.0	8.00	0.09	0.11
9	Kazookles	Other	1,206	93	7.7	0.25	0.85	0.10
10	Hair Toffee	Sugar	76	60	77.8	3.50	0.05	0.06
11	Fizzy Lifting Drinks	Sugar	79	47	60.0	2.25	0.06	0.05
12	Laffy Taffy	Sugar	54	33	62.3	1.24	0.04	0.04
13	SweeTARTS	Sugar	62	29	46.7	0.70	0.04	0.03
14	Nerds	Sugar	15	7	46.7	0.70	0.01	0.01
15	Fun Dip	Sugar	12	5	40.0	0.60	0.01	0.01

Table 1. Product profitability leaderboard (ranked by gross profit)

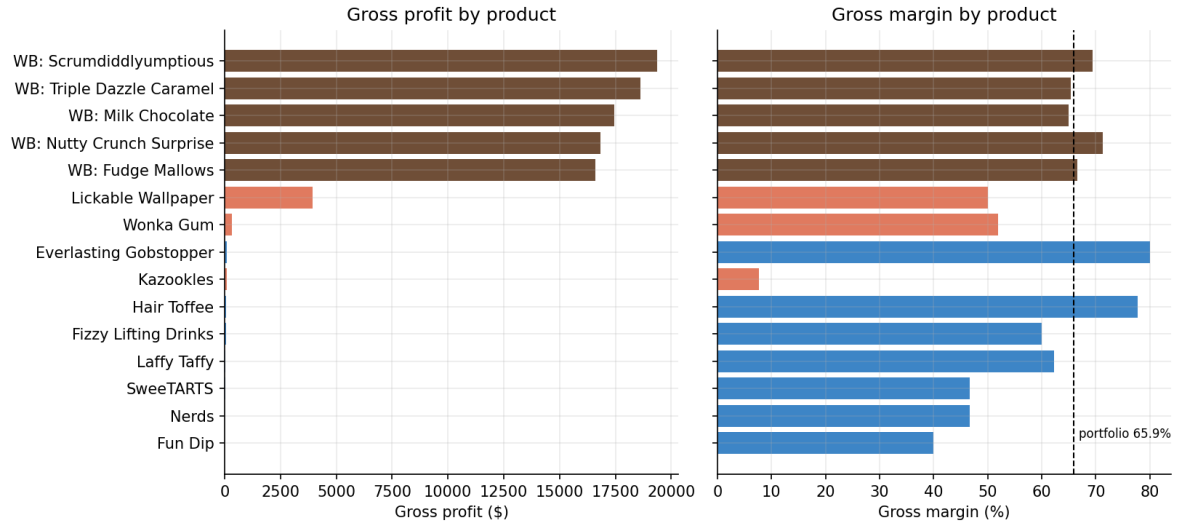


Figure 2. Gross profit and gross margin by product

4.3 Segmentation: high-sales, low-margin and low-sales, low-profit products

- **High-profit / high-margin (5 products):** all five Wonka Bars, 93% of revenue and 95% of profit.
- **High-sales / low-margin (1 product):** Lickable Wallpaper. It is the sixth-largest product with 5.5% of revenue, but a 50.0% margin means it delivers only 4.2% of profit. Its unit price is $20.00 against a unit cost of $10.00.
- **Low-sales / low-profit (5 products):** Kazookles, Wonka Gum, SweeTARTS, Nerds and Fun Dip. Each contributes at most 0.33% of profit.
- **Low-sales / high-margin (4 products):** Everlasting Gobstopper (80.0%), Hair Toffee (77.8%), Laffy Taffy (62.3%) and Fizzy Lifting Drinks (60.0%). Margins are healthy but volumes are tiny (3 to 10 order lines each).

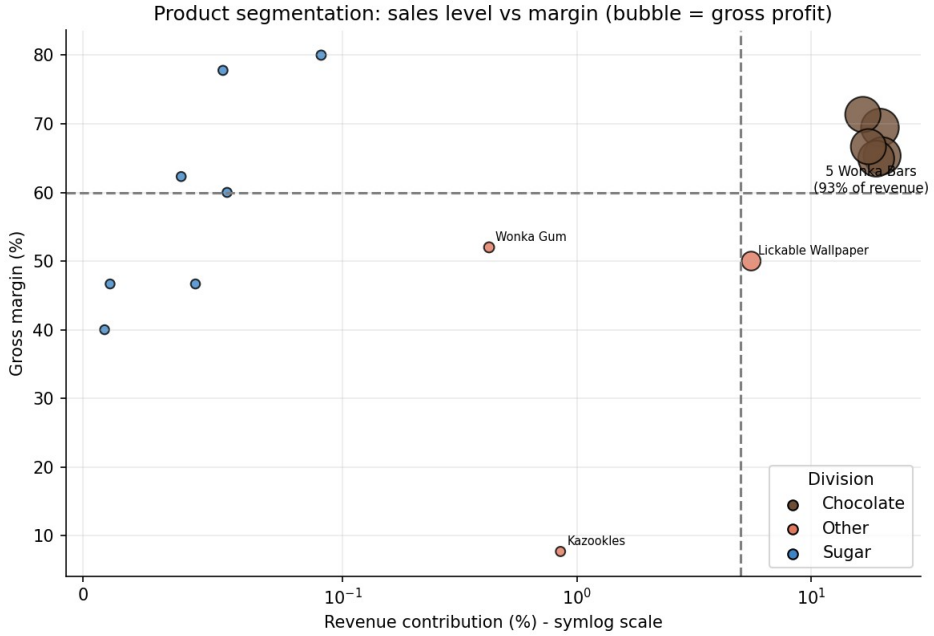


Figure 3. Product segmentation by revenue share and gross margin

4.4 Division-level performance

Division	Sales $	Profit $	GM %	Rev share %	Profit share %	Gap (pp)	Avg product GM %
Chocolate	131,693	88,825	67.4	92.88	95.06	+2.17	67.5

Division	Sales $	Profit $	GM %	Rev share %	Profit share %	Gap (pp)	Avg product GM %
Other	9,663	4,333	44.8	6.82	4.64	-2.18	36.6
Sugar	427	285	66.6	0.30	0.30	+0.00	59.1

Table 2. Division performance

Chocolate has both the largest scale and a margin (67.4%) above the portfolio average, and its profit share exceeds its revenue share by 2.2 percentage points. The Other division is the structural weak point: its 44.8% margin is 21 points below Chocolate, it gives back 2.2 points of profit share relative to revenue share, and its unweighted average product margin is only 36.6% because Kazookles (7.7%) pulls it down alongside Lickable Wallpaper (50.0%) and Wonka Gum (52.0%). Sugar's 66.6% margin looks strong, but it rests on only 40 order lines and $427 of sales, so it should not be treated as reliable.

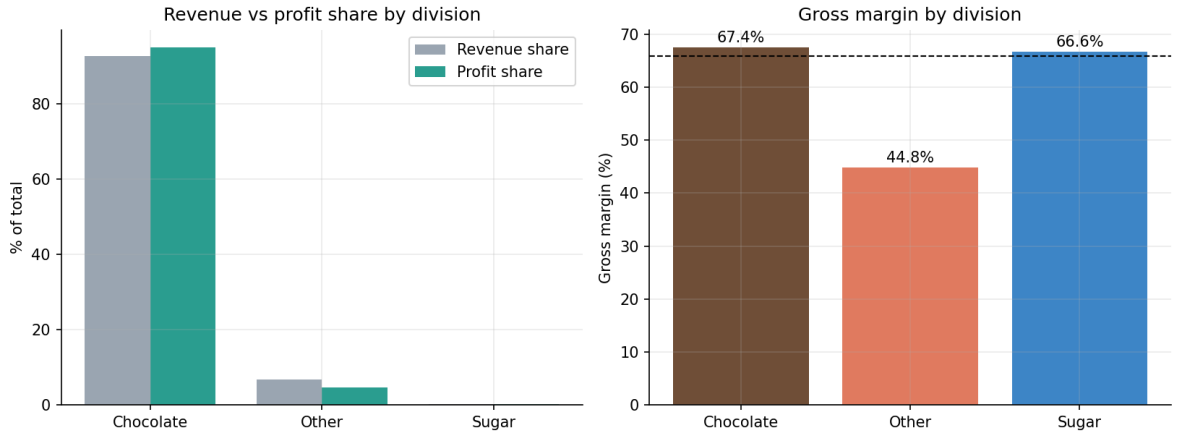


Figure 4. Revenue vs profit share and gross margin by division

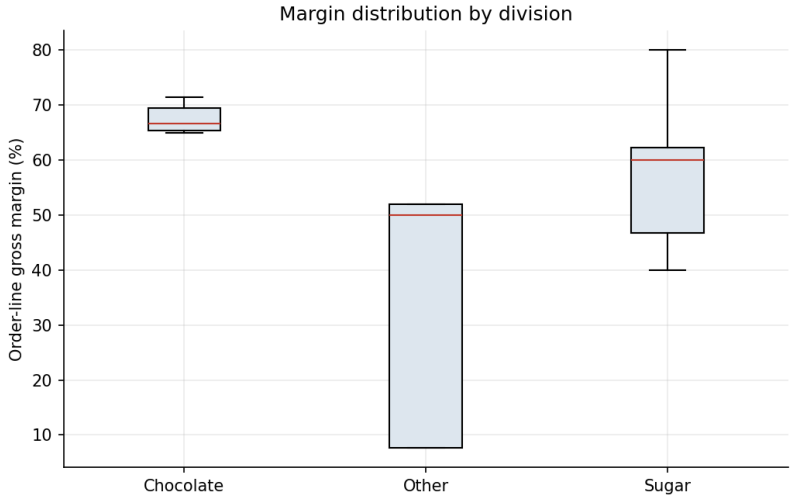


Figure 5. Distribution of order-line margins by division

4.5 Profit concentration (Pareto)

Only 5 of 15 products (33%) are needed to reach 80% of revenue, and the same 5 are needed for 80% of profit. The profit HHI is 1,831 at product level, but because all top products sit in one division and two factories, concentration is far higher at those levels (Table 3).

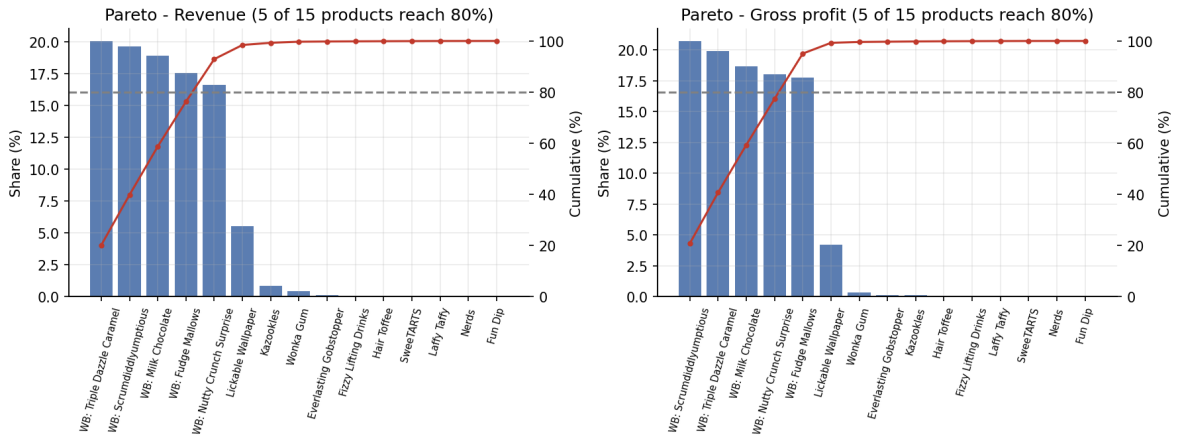


Figure 6. Pareto charts for revenue and gross profit

Dimension	Top item	Top share %	Top-3 share %	HHI	Risk
Product	Wonka Bar - Scrumdiddlyumptious	20.7	59.3	1,831	Low
Factory	Lot's O' Nuts	56.5	99.7	4,700	High
State/Province	California	19.8	40.3	765	Low
Region	Pacific	32.6	84.3	2,664	Moderate
Division	Chocolate	95.1	100.0	9,058	High

Table 3. Dependency indicators (shares of gross profit)

Two dependency risks stand out. First, the Chocolate division generates 95.1% of profit (HHI 9,058). Second, the factory Lot's O' Nuts produces 56.5% of profit and, together with Wicked Choccy's, 95.0%; a disruption at either site would put most of the business at risk. Geographic concentration is moderate: the Pacific region provides 32.6% of profit and California alone 19.8%; the top 5 states provide 50.7% and 16 of 59 states reach 80%. Regional margins are almost identical (65.5% to 66.4%), so geography influences volume but not margin.

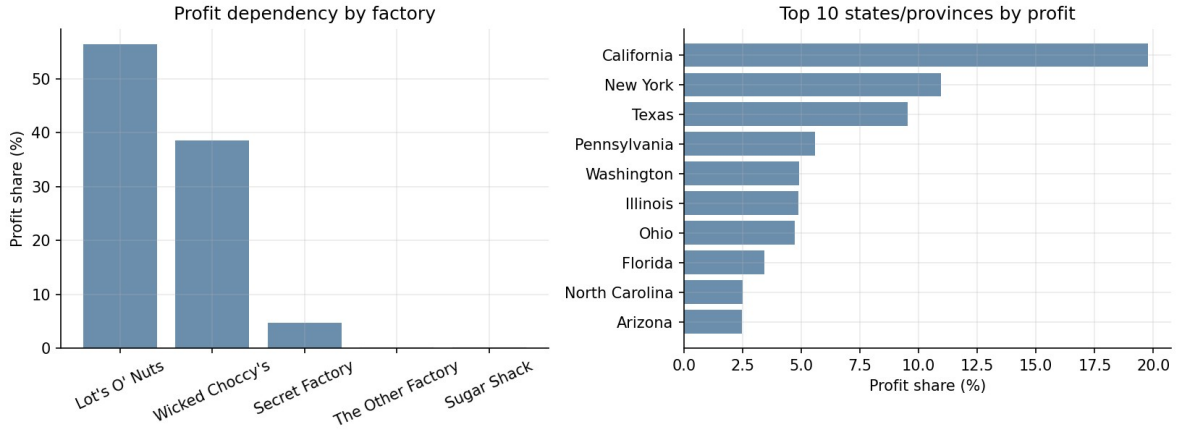


Figure 7. Profit dependency by factory and by state/province

Factory	Sales $	Profit $	GM %	Profit share %
Lot's O' Nuts	76,340	52,771	69.1	56.47
Wicked Choccy's	55,353	36,054	65.1	38.58
Secret Factory	8,588	4,345	50.6	4.65
The Other Factory	1,282	152	11.9	0.16
Sugar Shack	221	121	54.9	0.13

Table 4. Factory profitability (via product-to-factory mapping)

The Other Factory (Hair Toffee and Kazookles) earns an 11.9% margin and 0.16% of profit, driven by Kazookles. Secret Factory (Gobstopper, Lickable Wallpaper, Wonka Gum) earns 50.6%, well below the two chocolate factories.

4.6 Cost structure diagnostics

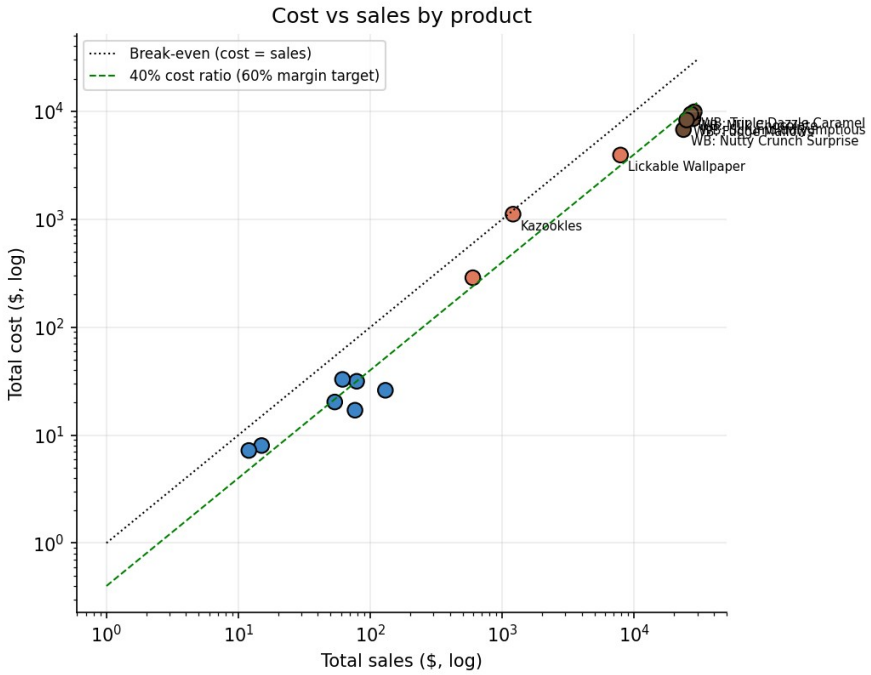


Figure 8. Cost vs sales; products above the dashed line miss the 60% margin target

On the cost-sales scatter, the five chocolate bars sit well below the 40% cost-ratio line, while Lickable Wallpaper (50% cost ratio) and Kazookles (92% cost ratio) sit above it. Unit economics explain why:



Figure 9. Unit economics: price split into cost and profit

Kazookles sells at $3.25 per unit and costs $3.00 to make, leaving $0.25 of profit per unit (compared with $2.11 to $2.50 for a Wonka Bar sold at a similar price). Lickable Wallpaper earns $10.00 profit per unit, the highest in the range, but its 50% margin is below target, so its cost base has the most room to improve in absolute terms.

Product	GM %	Cost ratio %	Rev share %	Flag
Lickable Wallpaper	50.0	50.0	5.54	Below target - cost renegotiation

Product	GM %	Cost ratio %	Rev share %	Flag
Wonka Gum	52.0	48.0	0.42	Discontinuation review
Everlasting Gobstopper	80.0	20.0	0.09	Low-volume watch
Kazookles	7.7	92.3	0.85	Critical - reprice & renegotiate cost
Hair Toffee	77.8	22.2	0.05	Low-volume watch
Fizzy Lifting Drinks	60.0	40.0	0.06	Low-volume watch
Laffy Taffy	62.3	37.7	0.04	Low-volume watch
SweeTARTS	46.7	53.3	0.04	Discontinuation review
Nerds	46.7	53.3	0.01	Discontinuation review
Fun Dip	40.0	60.0	0.01	Discontinuation review

Table 5. Margin-risk flags at the 60% threshold

4.7 Margin volatility

For every product, unit price and unit cost are constant across all order lines and months. Consequently the monthly gross margin of each product has a standard deviation of zero (Fun Dip has fewer than three months of data). This means that (a) no discounting, promotion or cost fluctuation is visible in the data, (b) portfolio margin moves only through product mix, and (c) margin risk is structural, built into list prices and standard costs, rather than caused by operational leakage. If actual transaction prices vary in practice, the dataset is not capturing it.

5. Recommendations

1. **Reprice and renegotiate Kazookles (priority 1).** At $3.25 price and $3.00 unit cost the product is close to break-even. Reaching a 60% margin needs either a price near $7.50 at current cost or a cost near $1.30 at current price; a realistic mix is a moderate price rise plus supplier or recipe renegotiation. If neither is feasible, discontinue. At constant volume, a 60% margin would add about $631 of profit.
2. **Renegotiate the cost of Lickable Wallpaper (priority 2).** It is the largest non-chocolate line, with 5.5% of revenue. A unit cost of $8.00 (-20%) or a price of $25.00 reaches 60% margin and would add about $786 of profit at constant volume.
3. **Review Wonka Gum, SweeTARTS, Nerds and Fun Dip for discontinuation.** All four have margins of 40% to 52% and under 0.5% of revenue each; the three Sugar items earn just $41 of profit combined. Decide whether to reprice, bundle or drop them to cut portfolio complexity.
4. **Test growth for high-margin niche products.** Everlasting Gobstopper (80%), Hair Toffee (77.8%), Laffy Taffy and Fizzy Lifting Drinks have healthy margins but almost no volume. Small promotions or placement tests could show whether demand can be grown.
5. **Protect the chocolate franchise and diversify production.** With 95% of profit in five bars made at two factories, supply continuity planning (backup co-manufacturing, safety stock) and incremental development of a second profit engine are the most valuable strategic actions.
6. **Use a margin threshold in pricing governance.** Adopt a floor (e.g. 60% gross margin for new SKUs) and review any product below it quarterly, using the dashboard's margin-threshold slider and risk flags.
7. **Fix the data.** Correct the ship-date anomaly, and capture actual transaction prices, discounts and period costs so margin volatility can be measured meaningfully.

Estimated impact: Items 1 and 2 together would add about $1,417 of gross profit over the 24-month period (+1.5% of the total) at constant volume; this ignores price elasticity and should be treated as an upper bound.

6. Limitations

- Gross profit uses manufacturing cost only; freight, marketing and overheads are excluded, so net profitability may differ.
- Constant unit price and cost per product suggest standardised or synthetic data; margin volatility and pricing-inefficiency conclusions are limited by this.
- Sugar-division products have very few observations (3 to 10 order lines each), so their margins are indicative only.
- Impact estimates assume unchanged volumes and no price elasticity.
- Ship dates are unreliable, so delivery and logistics analysis was out of scope.

7. Conclusion

Nassau Candy's profit comes from five Wonka Bars that deliver margins of 65% to 71%. Beyond them, the portfolio is thin: one meaningful product with a sub-target margin (Lickable Wallpaper), one product close to break-even (Kazookles), and a tail of items that barely register. The most actionable steps are repricing or recosting Kazookles and Lickable Wallpaper, rationalising the low-volume tail, and managing the concentration risk in the chocolate division and its two factories. The accompanying Streamlit dashboard lets management explore these results by date, division, margin threshold and product.

Appendix: Deliverables and reproducibility

- app.py: Streamlit dashboard (run with: streamlit run app.py).
- analysis.py: reproduces every table and figure in this paper into charts/ and outputs/.
- src/metrics.py: cleaning and KPI functions shared by both.
- outputs/: product, division, Pareto, dependency and factory tables as CSV; clean_data.csv.